CUU 1202 · Foundations of Critical and Creative Thinking

Strategies for Solving Problems and Making Decisions

Assignment 3 · Group 4 · [Presenter and group member names]

Problem solving and decision making

Problem solving

Finding a way to fix something that is not working, or to reach a goal we have not reached yet.

Decision making

Picking one option out of the choices we have, based on what matters most.

You cannot really do one without the other. To solve a problem you end up making a decision.

Steps in solving a problem

1. Define

Say clearly what the problem is.

2. Analyse

Collect facts and look for what is causing it.

3. Find options

Come up with as many solutions as we can.

4. Compare

Check each option against what we need.

5. Act

Pick one and put it into action.

6. Review

See if it worked and learn from it.

Strategies for solving problems

Strategy	What it is	When we use it
Algorithm	Follow a set of steps in order	When there is a known method, like a recipe
Trial and error	Try something, see if it works, try again	When there are few options and mistakes are cheap
Heuristic	A rule of thumb or shortcut	When we are short on time
Divide and conquer	Split a big problem into small parts	When the problem is big and complicated
Working backwards	Start from the goal and go back step by step	When we know the end result, like a deadline
Means-ends analysis	Keep reducing the gap between now and the goal	When we know the goal but not the route

Finding the root cause

5 Whys

Keep asking why until you reach the real cause. It usually takes about five times. It started at Toyota.

Fishbone diagram

Group the causes under headings like people, process, tools and environment.

Pareto (80/20)

A few causes make most of the trouble, so fix those first.

Creative ways to solve problems

Brainstorming

Write down as many ideas as possible. Judge them later.

Mind mapping

Put the problem in the middle and draw branches of ideas around it.

SCAMPER

Ask: can we substitute, combine, adapt, change, reuse, remove or reverse something?

Lateral thinking

Look at the problem from a different angle, an idea by Edward de Bono.

Six Thinking Hats

Look at it six ways: facts, feelings, risks, benefits, new ideas and process.

Analogy

Use a solution from a similar problem somewhere else.

Example: 5 Whys

Problem	Group assignments get submitted late.
Why?	The work started near the deadline.
Why?	Nobody was given a task.
Why?	There was no early meeting and no leader.
Why?	Everyone thought someone else would organise it.
Root cause	No clear roles and no responsibility.

Fix: give everyone a role in the first meeting, set small deadlines and agree on how to communicate.

How people make decisions

Rational model

List the options, decide what matters, score each option and pick the best one.

Satisficing

Herbert Simon said we rarely have enough time or information, so we pick the first option that is good enough.

Intuitive

Going with a gut feeling from experience. It works in familiar situations but can fail when we are new to it.

Recognition-primed

Experts match the situation to something they have seen before and act fast. This is from Gary Klein.

Tools that help us decide

Tool	What it does
Pros and cons list	Shows the good and bad sides of each option
Decision matrix	Gives each option a score based on what matters
Cost-benefit analysis	Compares what we spend with what we gain
SWOT analysis	Looks at strengths, weaknesses, opportunities and threats
Decision tree	Draws the choices and what could happen after each

Example: decision matrix

Our question: which tool should a student group use to work on a project? We score from 1 (bad) to 5 (very good).

What matters	Weight	WhatsApp	Google Docs	MS Teams
Easy to use	30%	5	4	3
Edit together at the same time	30%	1	5	4
Everyone sees messages fast	20%	5	3	4
Cost	20%	5	5	4
Total	100%	3.8	4.3	3.7

Answer: Google Docs wins. These scores are just an example, every group would choose its own.

Deciding as a group

Method	Good thing	Problem
Consensus	Everyone agrees, so everyone supports it	Takes a long time
Majority vote	Quick and fair	The people who lose may feel ignored
Leader decides	Fast and someone is responsible	Others may not get to give ideas
Nominal group technique	Everyone writes ideas alone first, then the group ranks them	Needs someone to lead the process

The best method depends on how important the decision is and how much time we have.

What can go wrong

Problem	What happens	What to do
Confirmation bias	We only look for proof that we are right	Look for evidence against our idea
Anchoring	We stick to the first idea or price we hear	Think of options before discussing
Sunk cost	We continue because we already invested	Only think about the future
Groupthink	Everyone agrees to avoid conflict	Let someone play devil's advocate
Overconfidence	We trust ourselves too much	Check our assumptions with facts

Summary

1. Understand the real problem before looking for solutions.
2. Choose a strategy that suits the type of problem.
3. Decide using clear criteria, watch out for bias and check the results afterwards.

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